Dimensionality Reduction and Principal Component Analysis (PCA)

1. Introduction

In many real-world machine learning problems, datasets contain a large number of features. Such high-dimensional data can lead to increased computational cost, overfitting, and difficulty in visualization.

To overcome these issues, we use **Dimensionality Reduction** techniques. One of the most widely used dimensionality reduction techniques is **Principal Component Analysis (PCA)**.

2. Dimensionality Reduction

2.1 What is Dimensionality?

Dimensionality refers to the number of features (attributes) in a dataset.

Example:

- Dataset with Height, Weight, Age, BMI $\rightarrow$ 4 dimensions
- Image of size 28×28 pixels $\rightarrow$ 784 dimensions

2.2 What is Dimensionality Reduction?

Dimensionality Reduction is the process of reducing the number of input features while retaining most of the important information present in the data.

2.3 Why Dimensionality Reduction is Needed

- Reduces computational cost
- Prevents overfitting
- Removes redundant features
- Improves visualization
- Improves performance of machine learning models

3. Simple Motivating Example

Consider a student dataset:

Student	Height (cm)	Weight (kg)
A	150	50
B	160	55
C	170	60
D	180	65

Observation:

- Height and Weight increase together
- Both features provide similar information

Keeping both features is redundant. Dimensionality reduction combines them into a smaller set of informative features.

4. Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is an **unsupervised** dimensionality reduction technique.

4.1 Definition of PCA

> PCA transforms correlated features into a smaller number of uncorrelated features called **principal components**, such that maximum variance is preserved.

4.2 Key Ideas of PCA

- PCA finds new directions in the data
- These directions capture maximum variance
- Principal components are orthogonal to each other
- The first component captures the most information

5. Intuition Behind PCA

PCA asks a simple question:

> In which direction does the data vary the most?

That direction contains the maximum information and becomes the **First Principal Component (PC1)**.

Subsequent components capture remaining variance while being perpendicular to previous components.

6. Daily-Life Analogy

PCA can be compared to choosing the best camera angle to capture a tall building. The best angle captures the maximum details. Similarly, PCA finds the best directions to represent the data.

7. Principal Components

- PC1: Direction of maximum variance
- PC2: Direction of second maximum variance, orthogonal to PC1
- PC3, PC4, ...: Decreasing importance

Principal components are:

- Orthogonal
- Uncorrelated

8. Mathematical Foundation of PCA (Simplified)

8.1 Data Standardization

Before applying PCA, data must be standardized:

$$X_{scaled} = \frac{X - \mu}{\sigma}$$

This ensures that all features contribute equally.

8.2 Covariance Matrix

The covariance matrix measures how features vary together:

$$Cov(X, Y) = \frac{1}{n} \sum (X - \bar{X})(Y - \bar{Y})$$

Interpretation:

- Positive covariance: features increase together
- Negative covariance: one increases while the other decreases
- Zero covariance: no relationship

8.3 Eigenvalues and Eigenvectors

From the covariance matrix, PCA computes:

- Eigenvectors $\rightarrow$ directions of principal components
- Eigenvalues $\rightarrow$ amount of variance in those directions

Higher eigenvalue means more information.

8.4 Principal Component Formula

A principal component is a linear combination of original features:

$$PC_1 = w_1X_1 + w_2X_2 + \cdots + w_nX_n$$

Where w_i are weights obtained from eigenvectors.

8.5 Projection and Dimension Reduction

Data is projected onto selected components:

$$Z = XW$$

Where:

- X = original data
- W = selected eigenvectors
- Z = reduced dataset

9. Explained Variance

Explained Variance Ratio is defined as:

$$\text{Explained Variance Ratio} = \frac{\text{Eigenvalue of a component}}{\text{Sum of all eigenvalues}}$$

It indicates how much information each principal component retains.

10. How PCA Reduces Dimensionality

$$\text{Original Data} \rightarrow \text{Standardization} \rightarrow \text{Covariance Matrix}$$
$$\rightarrow \text{Eigenvalues \& Eigenvectors} \rightarrow \text{Select Top Components}$$
$$\rightarrow \text{Reduced Dataset}$$

11. Why PCA is Used Before Machine Learning Models

- Reduces feature space
- Improves training speed
- Reduces overfitting
- Improves generalization

12. Limitations of PCA

- PCA is a linear technique
- Principal components may lose interpretability
- PCA assumes variance represents importance

13. Exam-Oriented One-Line Definitions

- **Dimensionality Reduction**: Reducing the number of features while preserving important information.
- **PCA**: An unsupervised technique that converts correlated features into uncorrelated components capturing maximum variance.

14. Final Summary

> Principal Component Analysis reduces dimensionality by finding new features that capture maximum variance, thereby simplifying data while preserving important information.

1. Introduction

A Feedforward Neural Network (FNN) is the simplest form of an artificial neural network. In this network, information flows only in one direction — from input to output — without any loops or feedback connections.

Feedforward neural networks form the foundation of more complex deep learning models.

2. Why is it called Feedforward?

The term **feedforward** means that data moves forward through the network without going back to previous layers.

$$\text{Input Layer} \rightarrow \text{Hidden Layer(s)} \rightarrow \text{Output Layer}$$

There are:

- No cycles
- No memory of past inputs
- No backward connections during prediction

3. Real-Life Analogy

A feedforward neural network is like a factory assembly line:

- Raw materials enter the factory
- Each station processes the product
- Final product comes out

No station sends the product back to the previous station.

4. Structure of a Feedforward Neural Network

A feedforward neural network consists of three types of layers:

4.1 Input Layer

- Receives input features
- Does not perform computation
- Simply passes values forward

Example:

$$(\text{Study Hours}, \text{Attendance})$$

4.2 Hidden Layer(s)

- Performs computations
- Learns patterns from data
- Can be one or more layers

4.3 Output Layer

- Produces final prediction
- Depends on problem type

Example:

$$\text{Pass / Fail}$$

5. Neuron Working

Each neuron performs two main operations.

5.1 Weighted Sum

$$z = w_1x_1 + w_2x_2 + \cdots + w_nx_n + b$$

Where:

- x_i = input values
- w_i = weights
- b = bias

5.2 Activation Function

$$a = f(z)$$

The activation function decides how strongly the neuron should activate.

6. Need for Activation Functions

Without activation functions:

- The network behaves like a linear model
- Multiple layers become ineffective

Activation functions introduce **non-linearity**, enabling the network to learn complex relationships.

7. Common Activation Functions

7.1 ReLU (Rectified Linear Unit)

$$f(z) = \max(0, z)$$

- Most commonly used
- Used in hidden layers
- Fast computation

7.2 Sigmoid Function

$$f(z) = \frac{1}{1 + e^{-z}}$$

- Output range: 0 to 1
- Used in binary classification

7.3 Softmax Function

$$f(z_i) = \frac{e^{z_i}}{\sum e^z}$$

- Used in multi-class classification
- Outputs class probabilities

8. Forward Propagation

Forward propagation is the process by which input data flows through the network to produce an output.

Steps:

1. Input is passed to hidden layers
2. Weighted sums are computed
3. Activation functions are applied
4. Output layer produces prediction

9. Simple Example

Input:

$$(\text{Study Hours}, \text{Attendance})$$

Hidden layer learns the importance of each input.

Output:

$$\text{Pass / Fail}$$

The model learns patterns such as:

High study hours can compensate for slightly low attendance.

10. Advantages of Feedforward Neural Networks

- Simple architecture
- Easy to implement
- Fast computation
- Foundation for deep learning models

11. Limitations

- No memory of past inputs
- Cannot handle sequential data
- Not suitable for time-series problems

12. Feedforward vs Other Neural Networks

- Feedforward NN: No memory, simple prediction
- Recurrent NN (RNN): Handles sequences
- Convolutional NN (CNN): Handles images

13. One-Line Exam Definitions

- **Feedforward Neural Network**: A neural network in which data flows only from input to output without feedback connections.
- **Neuron**: A basic computational unit that performs weighted sum and activation.
- **Activation Function**: A function that introduces non-linearity into the network.

14. Final Summary

A Feedforward Neural Network processes data in a single forward direction, using layers of neurons and activation functions to learn patterns and make predictions.